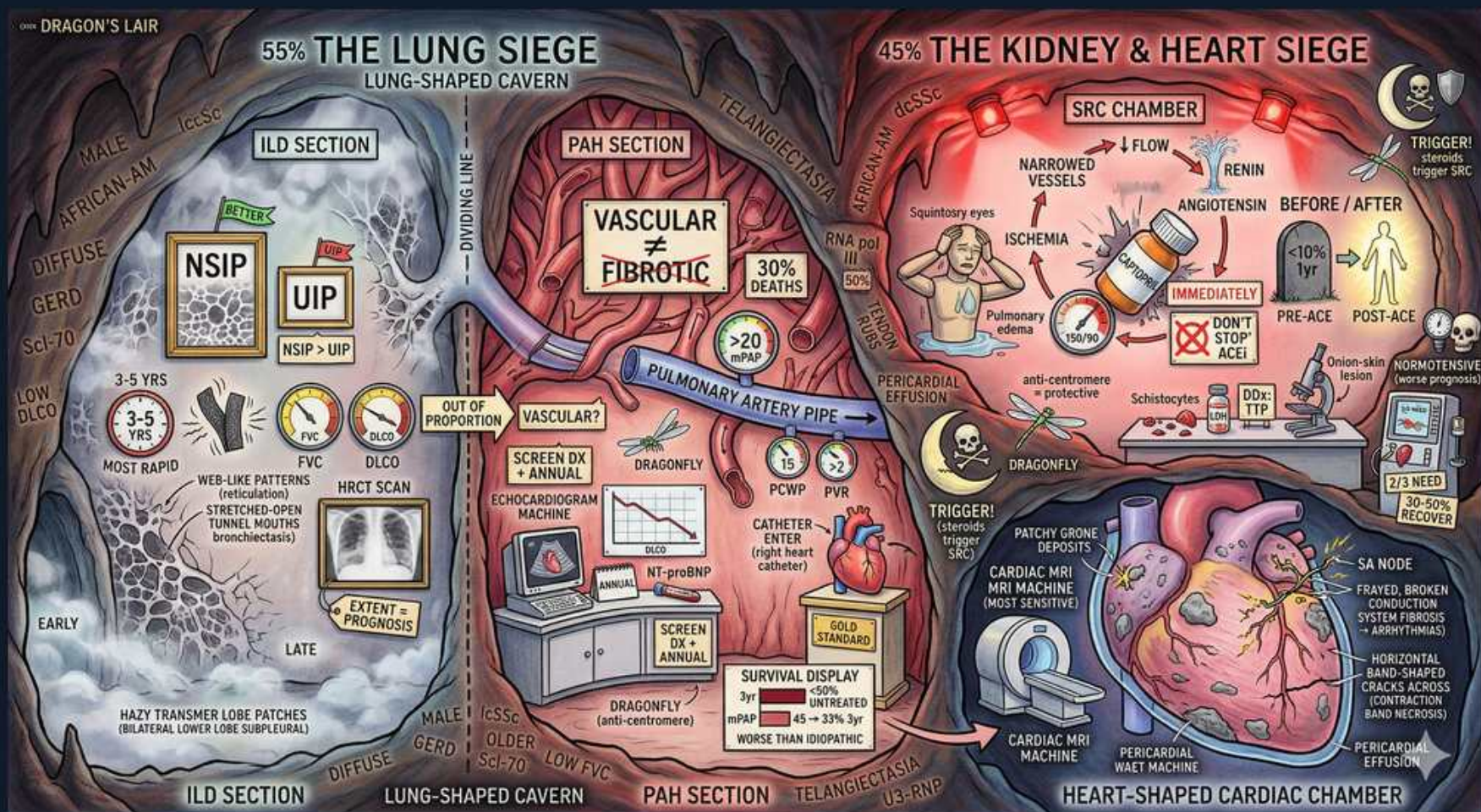


Scleroderma — Organ Involvement (Lung, Heart, Kidney)



1. ILD (interstitial lung disease)

WHAT YOU SEE

Lung-shaped cavern ILD section with NSIP/UIP boxes

WHAT IT ENCODES

ILD is now the leading cause of death in SSc (PAH is second), driven by alveolar fibroblast activation and collagen deposition. The NSIP (non-specific interstitial pneumonia) pattern predominates, showing ground-glass and lower-lobe subpleural reticulation; UIP (honeycombing) is less common. Highest risk with diffuse cutaneous disease and anti-Scl-70. Dyspnea on exertion and a dry cough are the typical first symptoms.

BOARD TRAP

A stem with SSc + lower-lobe fibrosis and you pick UIP/honeycomb pattern — WRONG. In SSc-ILD the NSIP pattern predominates (ground-glass > honeycombing), unlike RA-ILD and idiopathic pulmonary fibrosis where UIP dominates. Discriminator: NSIP is more cellular and more steroid/immunosuppression-responsive than UIP.

2. ILD screening (HRCT + PFTs)

WHAT YOU SEE

HRCT scan + FVC/DLCO dials in ILD section

WHAT IT ENCODES

Screen ALL newly diagnosed SSc patients with baseline HRCT and full PFTs including DLCO. A falling FVC (restrictive pattern, FVC <70% or a >10% relative decline) and falling DLCO track progression and prognosis. Combine with the SPAR/GAP-type risk stratification; serial PFTs every 3-6 months in early diffuse disease.

BOARD TRAP

Choosing chest X-ray or spirometry alone to screen SSc lungs — WRONG. CXR misses early disease; HRCT is the sensitive screen and DLCO is the most sensitive PFT parameter. Discriminator: an isolated low DLCO with preserved FVC points to PAH rather than ILD.

3. Anti-Scl-70 -> ILD risk

WHAT YOU SEE

Diffuse/Scl-70 labels along left margin pointing into ILD cavern

WHAT IT ENCODES

Anti-Scl-70 (anti-topoisomerase I) plus the diffuse cutaneous subtype carry the highest ILD risk and the worst pulmonary prognosis. The mnemonic: Scl-70 = SCarred Lungs. These patients need aggressive baseline and serial lung surveillance. Anti-Scl-70 and anti-centromere are mutually exclusive.

BOARD TRAP

Assuming anti-centromere predicts ILD — WRONG. Anti-centromere (limited/CREST) predicts isolated PAH, while anti-Scl-70 (diffuse) predicts ILD. Discriminator: match the antibody to the dominant lung lesion — Scl-70→fibrosis (ILD), centromere→vasculopathy (PAH).

4. PAH (pulmonary arterial HTN)

WHAT YOU SEE

PAH section, VASCULAR > FIBROTIC, 30% deaths

WHAT IT ENCODES

SSc-PAH is a Group 1 pulmonary arterial hypertension from intimal proliferation and obliterative vasculopathy (vascular > fibrotic), accounting for ~30% of SSc deaths. Classically seen in LONG-STANDING LIMITED/CREST disease with anti-centromere antibodies. Hemodynamic definition: mean PAP >20 mmHg, PCWP ≤15, PVR >2 Wood units on right heart cath.

BOARD TRAP

Calling SSc-PAH a Group 2 (left-heart) or Group 3 (lung-disease) PH and treating with diuretics alone — WRONG. SSc-PAH is Group 1 arterial PH requiring PAH-specific vasodilators. Discriminator: PAH is more typical of limited (anti-centromere) SSc; ILD-associated PH is more typical of diffuse disease.

5. PAH screening (echo annual)

WHAT YOU SEE

Screen Dx annual echocardiogram + NT-proBNP

WHAT IT ENCODES

Screen annually with transthoracic echo (estimated RVSP/TR jet) plus NT-proBNP and DLCO (the DETECT algorithm). Suspect PAH when DLCO falls out of proportion to FVC (FVC%/DLCO% ratio >1.6-1.8) or when NT-proBNP rises. Early detection markedly improves survival because PAH is often silent until advanced.

BOARD TRAP

Reassuring a dyspneic limited-SSc patient because the FVC is normal — WRONG. An isolated, disproportionately low DLCO with preserved lung volumes is the classic flag for PAH, not ILD. Discriminator: DLCO down + FVC normal = think vascular (PAH); DLCO and FVC both down = think ILD.

6. Right heart catheterization (gold standard)

WHAT YOU SEE

Catheter / gold standard right heart cath display

WHAT IT ENCODES

Right heart catheterization is the diagnostic GOLD STANDARD for PAH, directly measuring mean PAP (>20 mmHg), PCWP (≤15 to confirm pre-capillary/arterial PH), and PVR (>2 WU). It is required before starting PAH-specific therapy and to distinguish Group 1 from Group 2 left-heart PH.

BOARD TRAP

Diagnosing and treating PAH off the echo alone — WRONG. Echo estimates pressures and screens but overestimates/underestimates; RHC confirms the diagnosis and the hemodynamic group. Discriminator: a normal PCWP on RHC confirms pre-capillary (arterial) PH vs. an elevated PCWP indicating left-heart disease.

7. Scleroderma renal crisis (SRC)

WHAT YOU SEE

SRC chamber: narrowed vessels, renin up, malignant HTN

WHAT IT ENCODES

SRC = abrupt-onset malignant hypertension (often >150/90 with new headaches/visual changes) plus rapidly progressive AKI, driven by renal arteriolar onion-skin vasculopathy that activates the renin-angiotensin axis. Labs show MAHA with schistocytes and thrombocytopenia (a TMA picture). Most common in EARLY DIFFUSE SSc (first 4 years), especially anti-RNA-polymerase III.

BOARD TRAP

Mistaking the schistocytes + thrombocytopenia + AKI of SRC for TTP and giving plasma exchange — WRONG. SRC is a renin-driven TMA treated with ACE inhibitors, not PLEX. Discriminator: SRC has accelerated hypertension and a diffuse-SSc context with anti-RNA-pol III; TTP has ADAMTS13 deficiency and neuro signs without the malignant HTN/skin disease.

8. ACE inhibitor for SRC

WHAT YOU SEE

ACE inhibitor vial, DON'T STOP ACE warning

WHAT IT ENCODES

Short-acting ACE inhibitors (captopril, titrated rapidly, e.g. starting 6.25-12.5 mg and uptitrated every few hours) are first-line and LIFESAVING in SRC. Goal is controlled BP reduction; continue even as creatinine rises and even if temporary dialysis is needed, because renal recovery can occur over months. ACEi use cut SRC mortality from near-universal to a minority.

BOARD TRAP

Stopping the ACE inhibitor because the creatinine climbs after starting it — WRONG. A rising creatinine is expected and you DO NOT stop the ACEi; it is the treatment, not the cause. Discriminator: in SRC the benefit of continued RAAS blockade outweighs the transient creatinine bump — the only adjustment is supportive dialysis if needed, never ACEi withdrawal.

9. Steroids precipitate SRC

WHAT YOU SEE

Pre-ACE / post-ACE timeline with steroid trigger warning

WHAT IT ENCODES

High-dose corticosteroids (≥ 15 mg/day prednisone) are a well-established TRIGGER of scleroderma renal crisis, so they are avoided in early diffuse SSc. Other risk factors: anti-RNA-pol III, rapidly progressive skin thickening, and new anemia/pericardial effusion. If steroids are unavoidable, monitor BP and renal function closely.

BOARD TRAP

Reaching for high-dose prednisone to treat tight, progressing diffuse skin disease — WRONG. Steroids precipitate renal crisis in diffuse SSc. Discriminator: unlike most rheumatologic diseases where steroids are the workhorse, in scleroderma high-dose steroids are contraindicated/avoided — use methotrexate or mycophenolate for skin instead.

10. Cardiac fibrosis / conduction

WHAT YOU SEE

Heart-shaped cardiac chamber, patchy fibrosis, SA/conduction nodes

WHAT IT ENCODES

Primary myocardial involvement is patchy fibrosis from repetitive microvascular ischemia (Raynaud of the heart), causing arrhythmias, conduction blocks, and diastolic dysfunction (restrictive physiology). Often clinically silent but a poor prognostic marker; can progress to heart failure. Distinct from secondary cardiac disease due to PAH or SRC.

BOARD TRAP

Attributing all SSc cardiac disease to PAH-driven right heart strain — WRONG. SSc causes direct patchy myocardial fibrosis independent of pulmonary pressures. Discriminator: primary cardiac fibrosis shows conduction disease/diastolic dysfunction on a patchy, non-coronary distribution, best seen as late gadolinium enhancement on cardiac MRI.

11. Cardiac MRI (most sensitive)

WHAT YOU SEE

Cardiac MRI machine labeled most sensitive

WHAT IT ENCODES

Cardiac MRI with late gadolinium enhancement and T1/T2 mapping is the most sensitive test to detect early myocardial fibrosis and edema, before echo or symptoms appear. It distinguishes primary SSc cardiomyopathy from ischemic patterns and guides prognosis. Echo and ECG are screening adjuncts but less sensitive for fibrosis.

BOARD TRAP

Using a normal echo to rule out SSc cardiac involvement — WRONG. Echo misses patchy myocardial fibrosis that cardiac MRI detects. Discriminator: non-ischemic, patchy mid-wall late gadolinium enhancement on MRI is the signature of primary SSc cardiac fibrosis vs. subendocardial enhancement in coronary disease.

12. Pericardial effusion

WHAT YOU SEE

Pericardial wall machine / effusion in cardiac chamber

WHAT IT ENCODES

Pericardial involvement (effusion, occasionally pericarditis) occurs in SSc and is usually small and asymptomatic. Importantly, a new or moderate pericardial effusion in early diffuse disease can HERALD impending scleroderma renal crisis. Large effusions/tamponade are rare but carry a poor prognosis.

BOARD TRAP

Treating a new pericardial effusion in early diffuse SSc with high-dose steroids/NSAIDs and ignoring the kidneys — WRONG. The effusion may be an early warning of renal crisis, and steroids could precipitate it. Discriminator: in early diffuse SSc, a new effusion should trigger close BP/renal monitoring rather than reflexive steroid therapy.

13. ILD progression timeline (3-5 yrs)

WHAT YOU SEE

3-5 yrs clock, hazy lower-lobe patches (early->late)

WHAT IT ENCODES

ILD progresses fastest in the first 3-5 years of diffuse SSc, paralleling the window of maximal skin thickening — this is the critical surveillance and treatment window. Most lung function decline that ever occurs happens early, so aggressive monitoring (PFTs q3-6 months) and early immunosuppression in this period preserve the most lung.

BOARD TRAP

Deferring lung treatment until the patient is markedly symptomatic — WRONG. The greatest, often irreversible, FVC decline occurs early, so waiting loses salvageable lung. Discriminator: treat progressive ILD based on objective serial FVC/DLCO decline and HRCT extent, not on waiting for severe dyspnea.

14. Onset proximity to skin disease

WHAT YOU SEE

Onion-skin vascular / timing markers in PAH-ILD overlap

WHAT IT ENCODES

The highest-risk window for severe internal organ disease (ILD, renal crisis, cardiac) is the early diffuse phase when skin involvement is rapidly progressing. The onion-skin (concentric intimal hyperplasia) vasculopathy that drives renal crisis and PAH mirrors the active skin disease, so the rate/extent of skin thickening predicts visceral risk.

BOARD TRAP

Assuming a patient with longstanding stable limited skin disease shares the same renal-crisis risk as early diffuse disease — WRONG. Renal crisis clusters in EARLY DIFFUSE disease, not stable limited disease. Discriminator: tie organ risk to subtype and disease tempo — rapidly worsening diffuse skin = high renal/ILD risk; longstanding limited = PAH/GI risk instead.
